

A-7. Orchard Management Support				
Agriculture Land Conservation	High	High		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	Orchard management involves providing proper guidance and resources to optimize the growth and productivity of fruit orchards through irrigation management, soil conservation, pest control. Effective orchard management maximizes production, and sustains long-term profitability, while promoting environmental sustainability through practices like water conservation, soil fertility management, and biodiversity enhancement. Orchard management supports include organized tree planting, soil management, irrigation, pest control, pruning, and harvesting, as well as addressing challenges specific to the terrain, such as erosion in hilly areas. For the orchards in degraded hill slopes of the CHTs, establishing terraces along the natural contour of the slope is an effective way of preventing soil erosion by ensuring optimum water absorption and reducing run off. Drip irrigation is an effective method for irrigating orchards as it conveys irrigation water directly at the base of the plants hence maximizes water use efficiency. Moreover, regular pruning and tree training is essential for orchards in hilly areas to allow better sunlight penetration and air circulation among the trees. Planting windbreaks, such as tall grasses, shrubs, or trees along the perimeter of the orchard, reduces wind damage and protects the fruit trees from harsh weather conditions experienced in the elevated or sloped areas.								
	Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77
Applicable Sites									
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
			✓		✓		✓		
Feasibility criteria	- Assessing the soil for its fertility, texture, pH, and drainage properties and choosing suitable fruit tree species - Terracing or contour planting to reduce erosion risk, stabilize the soil, and enhance water retention. - Choosing tree species according to the climatic conditions of the area - Considering market access, value chain and demand of the forestry product as well as transportation and storage facilities.								
Operation and maintenance	Orchard management in hilly terrain require labor-intensive watering, pruning, pest control, and soil management. Additionally, long establishment period before								

	trees begin producing fruit require significant investment and patience from farmers, with risks from climate variability and potential pest infestations.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			

	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Fruit trees, especially deep-rooted species, help improve soil structure, reduce erosion, and enhance water retention in the soil. Their roots prevent soil from washing away during heavy rainfall, making them particularly valuable in areas prone to erosion and land degradation. • Integrating orchards with agro-ecological systems creates diverse habitats for wildlife, including birds, insects, and other beneficial organisms. This promotes biodiversity and can lead to healthier ecosystems that support pollinators and natural pest control agents. • Orchards contribute to better water retention and groundwater recharge through their root systems, improving water availability. • Orchards enhance nutrient cycling by shedding leaves and organic matter that enrich the soil. Their presence can reduce the need for external fertilizers, promoting more sustainable and organic farming practices.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			